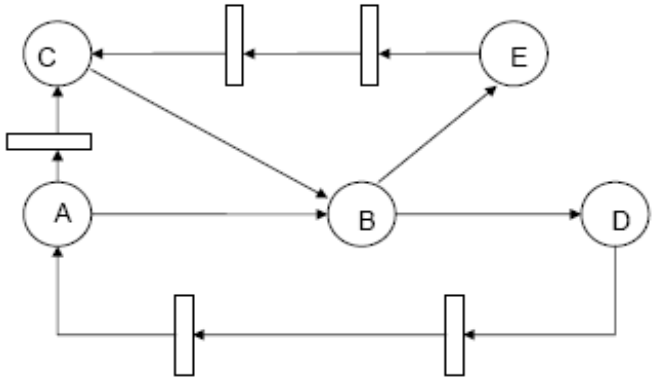
EC 551 Advanced Digital Design with Verilog and FPGAs

Fall 2026, Prof. Douglas Densmore

Homework #4

Due: 11/17

1. Consider the circuit shown below, with circles representing components and rectangles representing registers. Delays associated with circuit components are shown in the table below.



Component	A	B	C	D	E
Delay	22	10	18	13	12

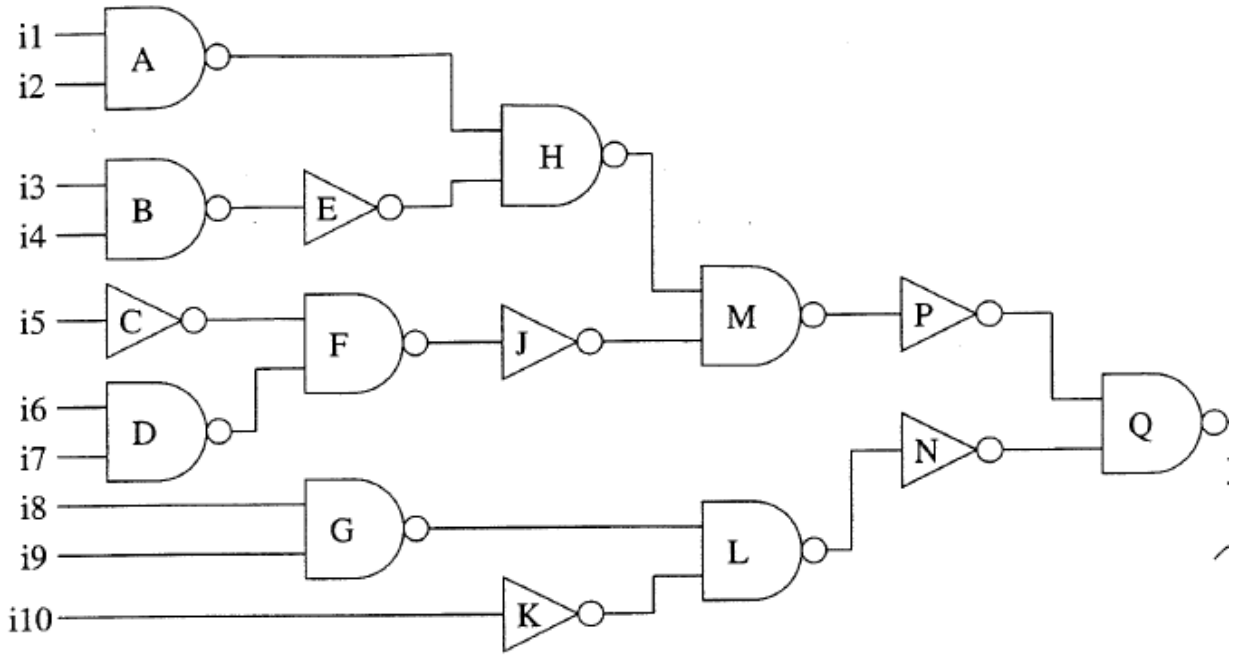
(a) Draw the retiming graph abstraction for the circuit shown above and compute the W and D matrices.

(b) Before retiming, what is the minimum clock period at which this circuit can be run?

(c) Is there a retiming that will allow this circuit to run with a clock period of 23? If so, find the retiming. If not, explain why there is no retiming.

Show all steps in your work, including deriving all "legal retiming" constraints and constraints on paths of length greater than 23, and drawing the resulting constraint graph (see lecture slides). If a retiming exists, draw the retimed circuit and write down the r(v) values for each component v.

2. Calculate the longest delay path from the output of each gate to the output of the circuit below. Do NOT include the gates' own delay in the calculation. What is this path in the circuit (list the gates that make up this path)? Is the critical path sensitizable? If so, give an input vector to do so. *Inverters have a delay of 1 unit and 2-input NANDS have a delay of 3 units. All wire delays are 1 unit.*



3. a. Find all the kernel and co-kernels of the following expressions. Label all the Level 0 kernels.

   F = af + bf + ag + cg + ade + bde + cde

   G = af + bf + ace + bce

   H = ade + cde

   b. Determine if there is a multiple-cube common subexpression of F, G, and H from the result of (a). If yes, represent F, G, and H using the common subexpression.